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TSV-Defect Modeling, Detection and Diagnosis Based on 3-D Full Wave Simulation and Parametric Measurement

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ABSTRACT In this paper, we present a TSV low-bandwidth equivalent lumped circuit model of common TSV defects as analyzed using a high-frequency structure simulator 3-D full wave simulation. By using the proposed model, the physical parameters of TSV are related to its electrical parameters, providing the possibility for defect diagnosis. The simulation results show that only some of the common defects tend to vary the electrical parameters of TSVs. Based on these conclusions, we present a novel TSV-defect detection and diagnosis method using switched-capacitor circuits. The proposed method can effectively detect significant void defects, pinhole defects, and missing micro-bump defects using parametric measurement and has been verified through HSPICE simulation.

INDEX TERMS 3D-ICs, TSV, defect modeling, test and diagnosis, switched capacitor.

I. INTRODUCTION

In the recent development of Integrated Circuit (IC) technology, delay and power consumption from interconnections have affected the system performance more significantly. To overcome interconnected scaling problems, the semiconductor industry has considered Through-Silicon Via (TSV) based 3-Dimensional (3-D) IC to be a promising option. In TSV based 3-D ICs, a TSV forms a vertical interconnect across dies. Since the 3-D technology is not mature, TSV defects can occur during TSV manufacturing or 3-D ICs bonding processes. Hence, it is necessary to implement a TSV test to increase the yields of TSVs and to identify defects to provide process feedback [1].

An accurate TSV-defect model is a precondition for TSV-defect detection and diagnosis. Although the TSV model has been widely studied [2]–[9], research on TSV-defect modeling has been limited. Kannan *et al.* [10] investigated multiple common TSV-defect models using structural analysis and formula derivation. Moreover, they proposed a test technique using a multi-tonal signal. By measuring the peak-to-average ratio, their method can detect common TSV defects with relatively high accuracy. Gong [11] modeled TSV voids, pinholes, misalignment micro-bumps, and missing micro-bumps defects through a High-Frequency Structure Simulator (HFSS) 3-D full wave simulation. The extracted TSV models were then used to conduct TSV-defect tests. Gong's work showed that the TSV parametric defect test requires a dedicated high-resolution measurement circuit due to minor variations in the parameters of defect TSV. Jung. *et al.* [12] have proposed a noninvasive TSV defect analysis method for high-speed TSV channel. In their paper, the author build the equivalent circuit models of high-speed TSV channel and defects. With designed and fabricated test vehicles, Jung's method is demonstrated with S-parameter and time-domain reflectometry measurement results. By measuring the S-parameter,open and short defects along the TSV channel can be distinguished and located without destroying the sample.

When an accurate TSV-defect model is built, defect tests and diagnosis can proceed. Currently, several TSV-defect testing and diagnosis methods exist. Taouil *et al.* [14] proposed a post-bond TSV test and diagnosis method for 3-D memory stacked on logic. In their study, a new memory-based interconnection test approach was used to develop fault models, their detection conditions, and the corresponding testing and diagnosis patterns. Deutsch and Chakrabarty [15] proposed a contactless pre-bond TSV test and diagnosis method using ring oscillators and multiple voltage levels. In their approach, TSVs are regarded as capacitive loads whose propagation delays can be measured by means of ring oscillators, allowing resistive open and leakage faults to be detected. Hsu *et al.* [16] proposed a built-in testing and diagnosis method for TSVs with various placement topologies. They presented a post-bond test and diagnosis approach for the TSV crosstalk fault. Based on their designed TSV grouping method, this approach enables a high-efficiency, low-area-overhead TSV test by reusing the existing boundary scan or IEEE 1500 wrapper cells. Zhang and Agrawal [17] proposed a optimization technology, namely "SOS3", for pre-bond diagnosis of TSV defects. Their technology is built on a recently proposed pre-bond TSV probing procedure. By using a three-stage optimization method named "SOS3", this technology can greatly speed up the pre-bond TSV test. Deutsch and Chakrabarty [18] have proposed a contactless pre-bond TSV fault diagnosis method based on duty-cycle detectors and Ring Oscillators (RO). In their article, the writer presents a method to create a regression model based on artificial neural networks to predict the fault size. The regression model can effectively determine whether a TSV has both leakage and resistive-open defects, thus increasing the resolution of detecting mixed TSV defects. In addition, Zhang *et al.* [19] proposed a low-cost TSV testing and diagnosis scheme based on the binary search method. This method was designed for the TSV probing test method, allowing the test efficiency to be significantly enhanced.

Although the aforementioned methods can accurately detect common TSV defects, they cannot provide detailed information on defects, which is useful for the TSV maker to improve the manufacturing technology. To solve this problem, two points need to be addressed. First, an accurate TSV-defect model is needed that can reflect the grades of defects in parameterization. Second, a high-resolution parametric fault test method is necessary. In this paper, we focus on both of these points.

First, we built a TSV-defect model using an HFSS 3-D full wave simulation in which the physical parameters of defects are linked to the electrical parameters of the extracted TSV-defect models. Second, we present a novel TSV-defect testing method based on the measured electrical parameters using Switched-Capacitor (SC) circuits. Based on the measurement results, we can diagnose the types, sizes, and locations of TSV defects. The remainder of this paper is organized as follows. In Section II, the TSV-defect models are modeled through the HFSS simulation. In Section III, the TSV parameter measurement and defect diagnosis mechanism is presented, and the simulation results are discussed and analyzed to verify the effectiveness of the proposed method. Finally, in Section IV, the conclusions are presented.

II. TSV DEFECT MODELS

In 3D ICs, the TSV is a vertical interconnection between upper dies and bottom dies. The shapes of the TSV include: Cylinder, Cone, Coaxial, etc. in which Cylinder TSV is the most common type. To insulate the TSV from the substrate, there is a dielectric layer surrounded the TSV. This unique structure makes TSV having very small resistance but relatively large capacitance compared with a traditional transmission line. Since TSV technology is not mature, new type defects may occur during TSV manufacturing process or 3D IC bonding process. To detect and diagnose these TSV defects, accurate TSV-defect models are necessary.

Figure 1 shows four common TSV defects as observed in SEM photos [13]. The first type of defect is the TSV micro-void defect, which is caused by insufficient copper filling during the TSV manufacturing process. A micro-void defect can increase the resistance of the TSV and result in a delay fault. If the defect is severe enough to cut off the TSV path, then signal transmission is compromised, which leads to an open fault. The second type of defect is the TSV leakage defect, which may be caused by imperfect oxide deposition during the TSV manufacturing process. When a leakage defect occurs, a conductive path forms between the TSV and the substrate, thus increasing the conductance of the TSV. The third and fourth types of defects are the misaligned micro-bump defect and the missing micro-bump defect, respectively, which may occur during the 3-D IC bonding process. Both defects result in a high resistance or even a completely open circuit at the bottom of the TSV, thus hindering signal transmission.

To analyze the impact of the common TSV defects and identify a suitable detection and diagnosis method, we use the HFSS to build the TSV-defect models. Figure 2 shows the implemented model of a fault-free TSV within a silicon substrate surrounded by an air-filled box. The physical parameters of different parts of the model are listed in TABLE 1. As shown in Fig. 2, the electric field is uniformly distributed along the fault-free TSV, reflecting experimental findings. Then, we use the HFSS to generate a low-bandwidth lumped circuit model, as shown in Fig. 3, which can represent the TSV in parameterization. In this figure, the lumped circuit model consists of one inductance L, two capacitances C_1 and C_2, and five resistances R_1, R_2, R_3, R_4, and R_{TSV}. Each component corresponds to a certain physical aspect of the TSV.

The inductance L and resistance R_{TSV} in the series between TSV port 1 and port 2 depend on the shape, size, and material of the TSV. For instance, as shown in Fig. 3, a cylindrical TSV with a 50 μm height and a 2.5 μm radius made of copper has an inductance of 0.026 nH and a resistance of 0.046 Ω. However, a cylindrical TSV with the same size made of graphite has an inductance of 0.227 nH and a resistance of 37.315 Ω.

The resistances R_1, R_2, R_3, and R_4 and capacitance C_1 and C_2 depend on the thickness and material of the dielectric and the bulk conductivity of the substrate. In the lumped model shown in Fig. 3, the dielectric has a thickness of 0.5 μm and a radius of 3 μm, while the substrate has a bulk conductivity of 10 siemens per meter. These parameters result in values of $R_1 = R_2 =$ 1e-5 Ω, $R_3 = R_4 = 452\ M\Omega$, and $C_1 = C_2 = 62\ fF$. If we decrease the thickness of the dielectric from 0.5 μm to 0.2 μm and keep all the other parameters unchanged, the resistances of R_3 and R_4 decrease to 39 $M\Omega$

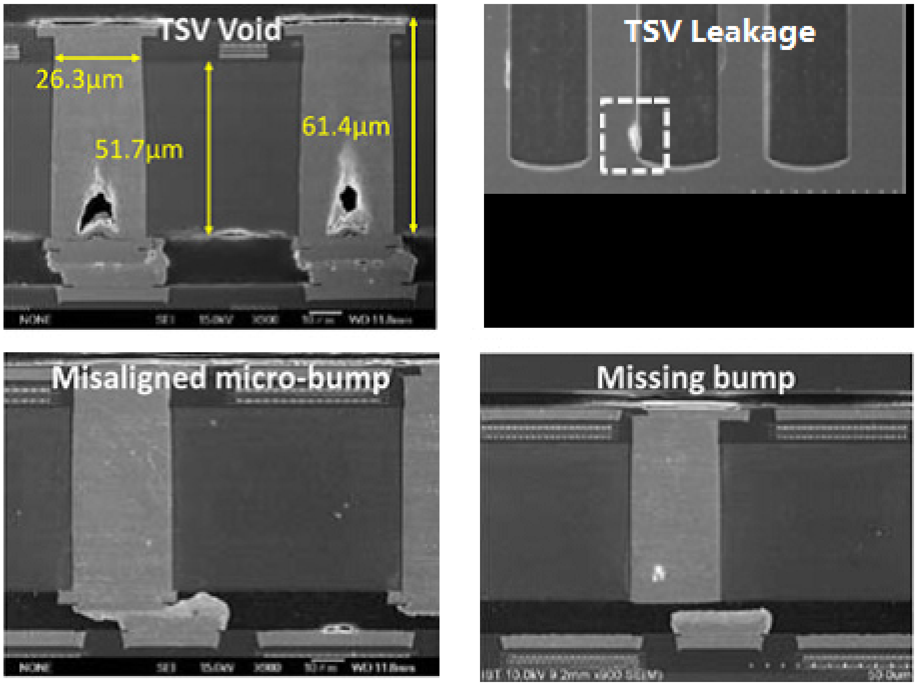

FIGURE 1. Four common TSV defects observed in SEM photos [13].

TABLE 1. Parameters of the implemented TSV structure.

Name	Material	Parameters
Cylindrical TSV	Copper	Height = 50 μm Radius = 2.5 μm
Dielectric	Silicon Dioxide	Thickness = 0.5 μm Radius = 3 μm
Substrate	Silicon	Length = 40 μm Width = 40 μm Height = 50 μm

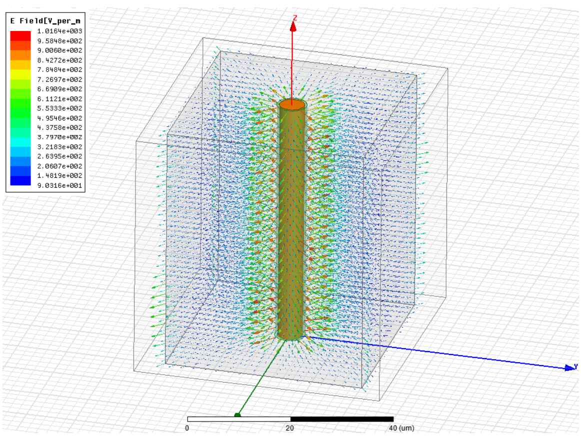

FIGURE 2. The Fault-free TSV model built in HFSS environment.

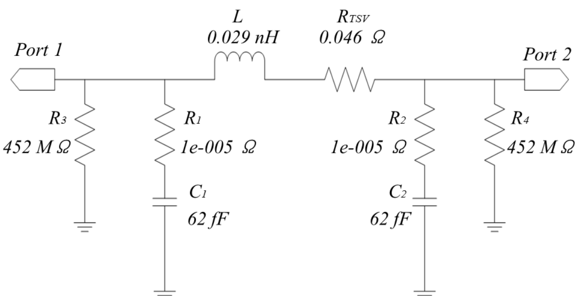

FIGURE 3. Low-bandwidth lumped circuit model of a fault-free TSV.

and the capacitances of C_1 and C_2 increase to 146 fF. This is because a thinner dielectric suffers from worse electric insulativity, causing the resistance between the TSV and substrate to increase. Since the dielectric is also equivalent to a capacitor whose two plates are the surface of the TSV and the surrounding of the substrate, a thinner dielectric corresponds to a shorter distance between the two plates, thereby leading to a larger capacitance.

All the above simulation data prove that the proposed low-bandwidth lumped circuit model conforms to the practical situation and can be used to reflect the variation of the TSV physical parameters. Since the TSV defects shown in Fig. 1 tend to vary the TSV physical parameters, we can simulate these defects in the HFSS and record the variation in the electric parameters caused by the different defects. This information is the precondition for the following detection and diagnosis of detects.

A. TSV VOID DEFECT

To study the impact of a TSV void defect, we insert different sizes of voids into the TSV and implement the

TABLE 2. Results of void defect simulation for various void sizes.

Cylindrical Void Size	$R_1(\Omega)$	$R_2(\Omega)$	$R_3(M\Omega)$	$R_4(M\Omega)$	$R_{TSV}(\Omega)$	$L(nH)$	$C_1(fF)$	$C_2(fF)$
Height = 1 μm Radius = 0.5 μm	1e-5	1e-5	258	258	0.046	0.029	62	62
Height = 1 μm Radius = 1 μm	1e-5	1e-5	546	546	0.047	0.029	62	62
Height = 1 μm Radius = 1.5 μm	1e-5	1e-5	203	203	0.048	0.029	62	62
Height = 1 μm Radius = 2 μm	1e-5	1e-5	291	291	0.049	0.029	62	62
Height = 1 μm Radius = 2.49 μm	1e-5	1e-5	126	126	0.050	0.029	62	62

HFSS simulation. For simplicity, the created voids are located at the middle of the TSV with cylindrical shapes. The void height is set to 1 μm and the radius of the different voids range from 0.5 μm to 2.49 μm.

Table 2 shows the parameter variation of the lumped circuit model under different void sizes. The resistances R_3 and R_4, which represent the resistance of the dielectric, exhibit random variation in the various situations. This variation may arise from the different mesh initializations when the HFSS employs different sizes of voids. Although R_3 and R_4 vary under different void sizes, they are always larger than 100 $M\Omega$, which indicates that a void defect does not decrease the electric insulativity of a TSV. Another concern is that R_{TSV}, which represents the resistance of the TSV, increases slightly with increasing void size. For a cylindrical void with a 1 μm height and a 0.5 μm radius, the R_{TSV} is equal to 0.046 Ω; For a cylindrical void with a 1 μm height and 2.49 μm radius, the R_{TSV} is equal to 0.050 Ω. This conclusion seems unlikely since a cylindrical void with a 2.49 μm radius nearly cuts off the TSV with 2.5 μm radius. In this case, the 0.050 Ω value seems too small because the void defect leads to an nearly open fault. To address this issue, we use the classical DC resistance equation to calculate the TSV resistance under different sizes of voids. The TSV resistance based on the classical DC equation satisfies the following formula:

$$R_{TSV} = \frac{\rho \cdot length}{\pi \cdot radius^2} \tag{1}$$

in which the *length* and *radius* are the height and radius of TSV, respectively, and ρ is the resistivity of the TSV.

Considering that the resistivity of copper is $1.75 \times 10^{-8}\ \Omega \cdot m$, according to TABLE 1, a fault-free TSV with 50 μm height and 2.5 μm radius has a resistance of 0.045 Ω, which basically conforms to the simulation results obtained in Fig. 3. To calculate the resistance of a TSV with a cylindrical void defect, we divide the TSV into three parts, as shown in Fig. 4. We then calculate the resistances of the three parts and add them together to obtain the lumped R_{TSV}. Assuming that the void defect is located in the middle of the TSV, the resistance of part 1 is expected to equal the resistance of part 3, which is 0.022 Ω based on formula (1). The resistance of part 2 can be calculated by the following formula:

$$R_{part2} = \frac{\rho \cdot L_{void}}{\pi \cdot (R_{TSV}^2 - R_{void}^2)} \tag{2}$$

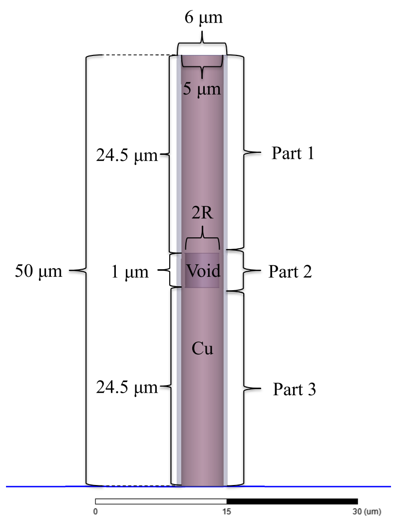

FIGURE 4. A cylindrical void defect located in the middle of a copper TSV.

in which L_{void} is the height of the cylindrical void, R_{TSV} is the radius of the TSV, and R_{void} is the radius of the cylindrical void.

Based on the data listed in TABLE 2, we calculate that for a cylindrical void with $L_{void} = 1\ \mu m$ and $R_{void} = 1\ \mu m$, the resistance of part 2 is equal to 0.001 Ω, thus resulting in $R_{TSV} = 0.045\ \Omega$. However, for a cylindrical void with $L_{void} = 1\ \mu m$ and $R_{void} = 2.49\ \mu m$, the resistance of part 2 is equal to 0.11 Ω, thus resulting in $R_{TSV} = 0.154\ \Omega$. These results prove that even a large void defect has little impact on the R_{TSV} parameter, which conforms to the HFSS simulation results.

Finally, we increase R_{void} to 2.5 μm to completely separate the TSV into two sections. The HFSS simulation shows that the low-bandwidth lumped circuit model can be transformed to the topology shown in Fig. 5. Compared with fault-free case, the small resistor and inductor connecting the two ports of the TSV are replaced by two small capacitors C_3 and C_4. This is because the open defect cut off the TSV to form two cross-sections, which are the two plates of new formed capacitors.

Another factor is how the location of such a void defect affects the electrical parameters. To address this issue, we conducted another simulation in which we maintained the size of the void at 1 μm height, 1 μm radius unchanged and swept the void location from top to bottom. The simulation results are shown in TABLE 3, in which a location of 10%

TABLE 3. Simulation results of an entirely open fault caused by a void defect for various void locations.

Void Location	$R_1(\Omega)$	$R_2(\Omega)$	$R_3(M\Omega)$	$R_4(M\Omega)$	$C_1(fF)$	$C_2(fF)$	$C_3(fF)$	$C_4(fF)$
10%	1e-5	1e-5	3150	198	5.4	55.9	0.3	0.3
30%	1e-5	1e-5	478	517	18.4	43	0.3	0.3
50%	1e-5	1e-5	714	2209	30.1	31.4	0.3	0.3
70%	1e-5	1e-5	640	614	42.4	18.9	0.3	0.3
90%	1e-5	1e-5	195	2044	55.3	6.0	0.3	0.3

TABLE 4. Simulation results of the TSV with a pinhole defect for various pinhole sizes.

Pin-hole Area	$R_1(\Omega)$	$R_2(\Omega)$	$R_3(\Omega)$	$R_4(\Omega)$	$R_{TSV}(\Omega)$	$L(nH)$	$C_1(fF)$	$C_2(fF)$
$1\ \mu m \times 1\ \mu m$	1e-5	1e-5	30396	30396	0.046	0.029	58	58
$2\ \mu m \times 2\ \mu m$	1e-5	1e-5	13710	13710	0.046	0.029	55	55
$3\ \mu m \times 3\ \mu m$	1e-5	1e-5	8237	8237	0.046	0.029	51	51
$4\ \mu m \times 4\ \mu m$	1e-5	1e-5	5878	5878	0.046	0.029	48	48
$5\ \mu m \times 5\ \mu m$	1e-5	1e-5	4291	4291	0.046	0.029	44	44

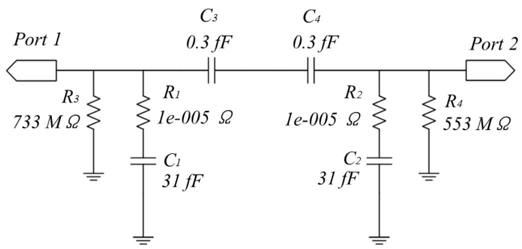

FIGURE 5. The lumped circuit model for a TSV with open fault.

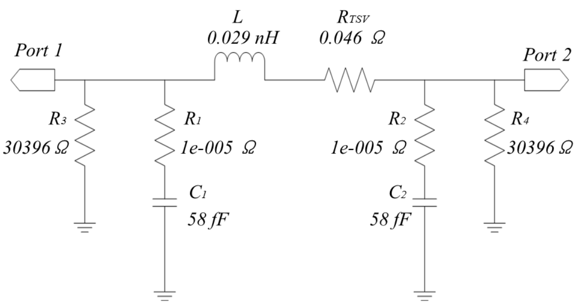

FIGURE 6. The lumped circuit model for $1\ \mu m \times 1\ \mu m$ pinhole defect.

means that the void defect is located at 10% of the TSV length from port 1. The data in TABLE 3 show that the location of an open fault caused by a void defect is found to the capacitances C_1 and C_2. Assuming that the total capacitance of TSV is C, a void defect located at position x divides this capacitance into xC and $(1-x)C$. For instance, as shown in TABLE 3, the total capacitance of TSV is approximately $61.3\ fF$. A open fault located at the 30% position results in values of $C_1 = 30\% \times 61.3\ fF = 18.39\ fF$ and $C_2 = (1-30\%) \times 61.3\ fF = 42.91\ fF$, which basically conforms to the simulation results.

In conclusion, the void defect has two main characteristics: First, small or medium void defects that do not completely cut off the TSV only minimally affect the TSV electric parameters and hence cannot be detected by parameter measurement. Second, large void defects that completely cut off the TSV lead to open faults and change the topological structures of the lumped circuit model. In such a case, the distribution of the separated TSV capacitors depends on the location of void defect.

B. TSV PINHOLE DEFECT

To analyze the impact of a TSV pinhole defect, we introduce a square pinhole in the dielectric of the fault-free TSV model and simulate it in the HFSS environment. Figure 6 shows the low-bandwidth lumped circuit model of a TSV with a $1\ \mu m \times 1\ \mu m$ pinhole defect. Compared with the fault-free case, the capacitances of C_1 and C_2 both decrease from $62\ fF$ to $58\ fF$, whereas the resistances of R_3 and R_4 decrease more significantly, both from $452\ M\Omega$ to $30396\ \Omega$. Generally, the TSV body is surrounded by a dielectric, which is typically made of silicon dioxide. In the fault-free case, the TSV body is insulated from the substrate which always connects to the ground, thus resulting in a high R_3 and R_4. When a pinhole defect occurs in the dielectric, the dielectric is broken, and a leakage path forms between the TSV body and the substrate, thus decreasing R_3 and R_4. Furthermore, pinhole defect decreases the area of dielectric, thus leading to a smaller C_1 and C_2. To verify the above predictions, we performed another simulation with pinhole sizes ranging from $1\ \mu m \times 1\ \mu m$ to $5\ \mu m \times 5\ \mu m$. The simulation results, shown in TABLE 4, confirm that the resistances R_3 and R_4 and capacitor C_1 and C_2 decrease as the pinhole size increases.

Another concern is how the location of the pinhole defect affects the electric parameters of the TSV. To address this issue, we performed another simulation in which we kept the size of the pinhole constant at $1\ \mu m \times 1\ \mu m$ and moved the pinhole location from port 1 to port 2. The simulation results

TABLE 5. Simulation results of the TSV with a 1 $\mu m \times 1$ μm pinhole defect for various pinhole locations.

Pin-hole Location	$R_1(\Omega)$	$R_2(\Omega)$	$R_3(\Omega)$	$R_4(\Omega)$	$R_{TSV}(\Omega)$	$L(nH)$	$C_1(fF)$	$C_2(fF)$
10%	1e-5	1e-5	29273	29273	0.046	0.029	59	59
30%	1e-5	1e-5	27384	27384	0.046	0.029	58	58
50%	1e-5	1e-5	28007	28007	0.046	0.029	58	58
70%	1e-5	1e-5	29773	29772	0.046	0.029	59	59
90%	1e-5	1e-5	29406	29406	0.046	0.029	59	59

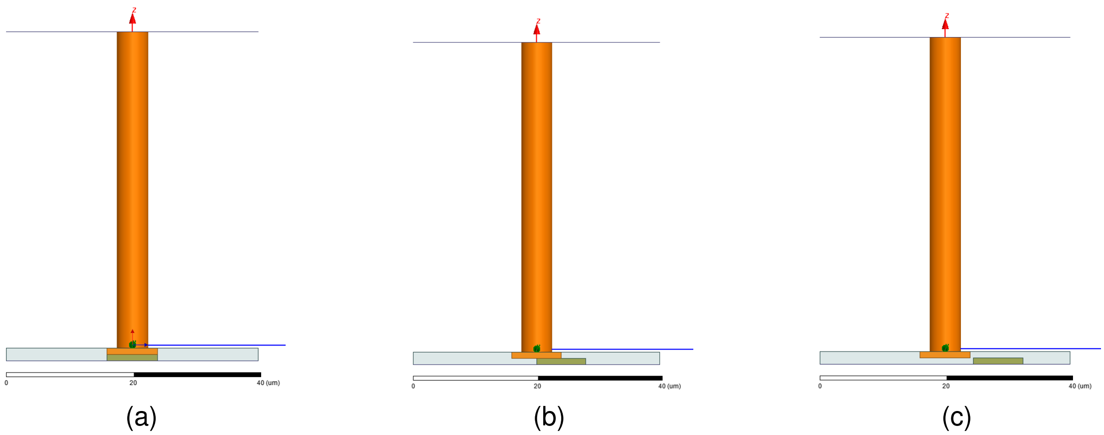

FIGURE 7. TSV defect model after die bonding process. (a). Fault-free TSV; (b). TSV with misaligned defect; (c). TSV with missing micro-bump defect.

TABLE 6. The simulation results of the TSV misaligned defect for various joint areas.

Bonding Area	$R_1(\Omega)$	$R_2(\Omega)$	$R_3(M\Omega)$	$R_4(M\Omega)$	$R_{TSV}(\Omega)$	$L(nH)$	$C_1(fF)$	$C_2(fF)$
100%	1e-5	1e-5	218	218	0.061	0.027	70	70
75%	1e-5	1e-5	129	129	0.065	0.028	70	70
50%	1e-5	1e-5	152	152	0.068	0.027	70	70
25%	1e-5	1e-5	247	247	0.072	0.023	70	70
2.5%	1e-5	1e-5	250	250	0.080	0.023	70	70

are shown in TABLE 5, in which the location of 10% means that the pinhole defect is located at 10% of the TSV length from port 1. We find that the location of a pinhole defect has little impact on the electric parameters of the TSV.

C. TSV MISALIGNED OR MISSING MICRO-BUMP DEFECT

During the TSV bonding process, the substrate is thinned to expose the buried TSV and then a micro-bump, usually made of copper, is connected to the exposed TSV for bonding purpose. The bonding bumps on the other die are usually made of Tin. When copper bumps are bonded to Tin bumps, TSV misaligned or missing micro-bump defect may occur. Figure 7 shows these defects modeled in the HFSS environments, in which the TSV in Fig. 7(a) is fault-free, the TSV in Fig. 7(b) has a misaligned defect, and the TSV in Fig. 7(c) has missing micro-bump defect. We simulated these three models and found that, for the fault-free TSV and the TSV with misaligned defect, the topological structures of low-bandwidth lumped circuits models are as same as the fault-free case shown in Fig. 3. For the TSV with missing micro-bump defect, the topological structure of low-bandwidth lumped circuits model is as same as the case in Fig. 5. This result brings us to think that a misaligned defect is equivalent to a void defect that yet completely cut off the TSV occured at TSV bottom whereas a missing micro-bump defect is equivalent to a void defect that completely cut off the TSV occured at TSV bottom. To prove this point, we implemented a simulation in which we decreases the bonding area between the copper bumps and the Tin bumps from 100% to 2.5%. The simulation results are shown in TABLE 6.

When the bonding area decreases, the TSV resistance increases. When the bonding area is 2.5%, which corresponds to a nearly open fault, the resistance of the TSV increases to 0.080 Ω, which is still relatively small. Such a small variation in R_{TSV} is hard to detect through the parameter measurement. When the bonding area becomes 0%, the low-bandwidth lumped circuits model changes to the topological structure shown in Fig. 5 with R_1 = 1e-5 Ω, R_2 = 1e-5 Ω, R_3 = 178 $M\Omega$, R_4 = 1031 $M\Omega$, C_1 = 67 fF,

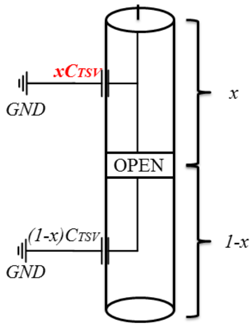

FIGURE 8. TSV open fault model.

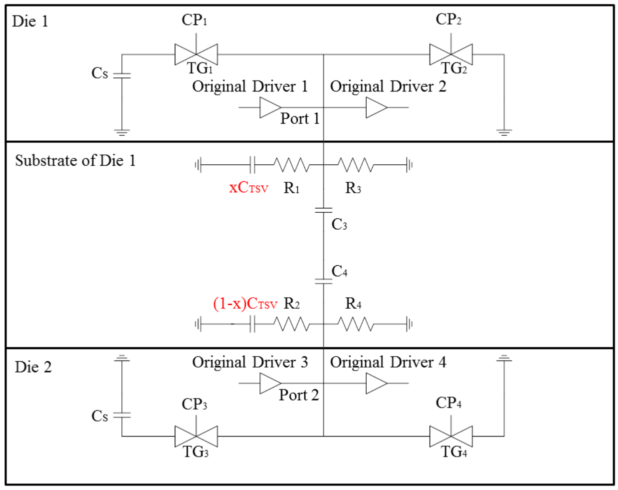

FIGURE 9. The proposed DfT circuits for pre-bond and post-bond TSV test.

$C_2 = 2.7\,fF$, $C_3 = 4.6\,fF$, and $C_4 = 4.6\,fF$. C_1 and C_2 vary significantly as an open fault exists at the end of the TSV. All these simulation results are similar as the case of void defect, from which we conclude that the TSV misaligned or missing micro-bump defects are equivalent to void defects occured at the end of the TSV.

As a conclusion, the TSV misaligned defect has little impact on the electrical parameters of low-bandwidth lumped circuit models, hence, it cannot be detected by parameter measurement. In contrast, the TSV missing micro-bump defect corresponds to an open fault at the bottom of the TSV, thereby significantly increasing the TSV resistance. Therefore, this TSV missing micro-bump defect can be detected by parameter measurement.

III. TSV DEFECT DIAGNOSIS BASED ON PARAMETERS MEASUREMENT

In the preceding section, we present the low-bandwidth lumped circuits models of several common TSV defects as investigated through HFSS simulation. The simulation results show that significant void defects, pinhole defects, and missing micro-bump defects can affect the electrical parameters of the TSV. In other words, these defects can be detected through parameter measurement. In the following section, we focus on detecting and diagnosing these three types of defects.

A. TSV VOID DEFECT DETECTION AND DIAGNOSIS

In Section II, we determined that a void defect that has not completely cut off the TSV has little impact on the electrical parameters of the lumped circuit models. To detect these kinds of defects, the X-ray method or sonar method may be useful. However, when the void defect is large enough to separate TSV and form an open fault, the electrical parameters are expected to vary significantly. The simulation results in TABLE 3 indicate that a TSV with an open fault has an abnormal C_1 and C_2, which depends on the fault location. For clarify, we schematize a TSV with an open fault using Fig. 8. Here, the TSV is divided into two separated parts beside the open fault, thus the TSV capacitor is divided into two independent capacitors. Assuming that the fault location is x, the capacitance of the top part is equal to xC_{TSV}, and the capacitance of the bottom part is equal to $(1-x)C_{TSV}$. When such a TSV is tested from the top port, only the capacitor of xC_{TSV} can act as a load. Conversely, when such a TSV is tested from the bottom port, only the capacitor of $(1-x)C_{TSV}$ can act as a load. Therefore, we can measure the value of xC_{TSV} and $(1-x)C_{TSV}$ to detect the open fault and diagnose the fault location.

To measure xC_{TSV} and $(1-x)C_{TSV}$, we propose a novel measurement method based on SC circuits. The main advantages of this method are its high accuracy, strong robustness, and the ability to be used in both pre-bond tests and post-bond tests.

The measurement circuit consists of four Transmission Gates (TGs) and two off-the-shelf capacitors, as shown in Fig. 9. The upper two TGs are located beside TSV port 1 and the bottom two TGs are located beside TSV port 2. For a TSV under test, the two TGs on the left are connected with the off-the-shelf capacitor C_s, and the two TGs on the right are connected to the ground. CP_1, CP_2, CP_3, and CP_4 are the four control signals for the four TGs. To reflect the practical situation, we also add four original tristate drivers, which belongs to the TSV functional circuits. During the whole testing process, the four original tristate drivers keep a high-Z state. Since these four drivers are not included in our DfT structures, the reason we put them here is to consider the parasitic capacitance they induced.

At the beginning of the test, the test initialization should be implemented to precharge C_s to V_{DD}. Then TG_1 and TG_2 are alternatively closed in the period of T_c, as shown in Fig. 10, while TG_3 and TG_4 remain open. The closed period of TG_1 is tp_1 seconds, during which xC_{TSV} is charged by C_s. The closed period of TG_2 is tp_2 seconds, during which the xC_{TSV} is discharged to ground. A duty-cycle scheme between t_{p1} and t_{p2} prevents a race hazard situation. Assuming that the initial voltage on C_s is V_{DD} and that the transferred charge during t_{p1} seconds is Q, the following relationship applies:

$$Q = \frac{xC_{TSV} \cdot C_s}{xC_{TSV} + C_s} \cdot V_{DD} \tag{3}$$

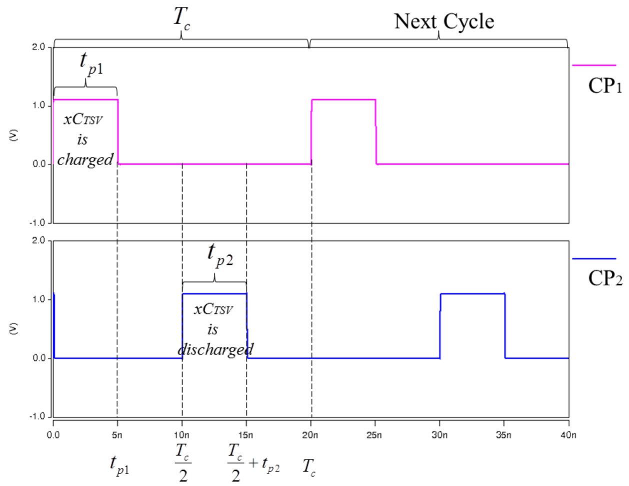

FIGURE 10. The time sequence of CP_1 and CP_2.

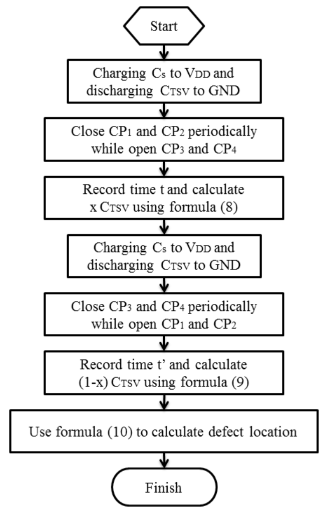

FIGURE 11. The test flow of TSV void defect detection and diagnosis.

These charges are transferred during T_c seconds, thus indicating that the average current I satisfies the following formula:

$$I = \frac{Q}{T_c} = \frac{xC_{TSV} \cdot C_s}{(xC_{TSV} + C_s) \cdot T_c} \cdot V_{DD} \tag{4}$$

If we set the charging time t_{p1} to be considerable large, the average voltage during the T_c period can be described by the following:

$$V = \frac{C_s}{xC_{TSV} + C_s} \cdot V_{DD} \tag{5}$$

In this case, the SC circuits are equivalent to a resistor with the following value:

$$R_{SC} = \frac{V}{I} = \frac{T_c}{xC_{TSV}} \tag{6}$$

According to the HFSS simulation results, xC_{TSV} is very small, typically on the order of dozens of fF. Thus, directly measuring xC_{TSV} is challenging to realize. Using SC circuits, we transform xC_{TSV} to an equivalent resistor R_{SC} whose value satisfies formula (6). R_{SC} is straightforward to measure since it forms a first-order discharging circuit with the capacitance C_s. When C_s is precharged to V_{DD}, the discharging process is described by the following formula:

$$V_{CS} = V_{DD} \cdot e^{-\frac{t}{\tau}} = V_{DD} \cdot e^{-\frac{t}{R_{SC} \cdot C_s}} \tag{7}$$

in which V_{CS} is the voltage across the capacitance C_s and τ is equal to $R_{SC} \cdot C_s$.

To measure xC_{TSV}, we insert formula (6) into formula (7) and record the discharging time t when V_{CS} drops to a certain threshold voltage V_{TH}, which follows the relationship:

$$xC_{TSV} = \frac{ln(\frac{V_{DD}}{V_{TH}}) \cdot C_s \cdot T_c}{t} \tag{8}$$

The aforementioned test process is simplely described by the first three steps in Fig. 11. All these steps can be implemented before die bonding since up to this point we do not need any support of the DfT structure in die 2. In other words, the proposed method can detect the existence of an open fault in the pre-bond stage.

To further conduct a defect diagnosis, we perform the test again after the die bonding process. In the post-bond stage, TG_3 and TG_4 are connected to port 2 of the TSV. This time, we target $(1-x)C_{TSV}$ using the mirrored test operation, during which we open TG_1 and the TG_2 and alternatively close TG_3 and TG_4 over a period of T_c. By recording the discharging time t' when V_{CS} drops to a certain threshold voltage V_{TH} in this mirrored test operation, the $(1-x)C_{TSV}$ can be calculated by formula:

$$(1-x)C_{TSV} = \frac{ln(\frac{V_{DD}}{V_{TH}}) \cdot C_s \cdot T_c}{t'} \tag{9}$$

Above test process is simplely described by the fourth to the sixth steps in Fig. 11.

Finally, to calculate the location x of open fault, we bring the calculated results of formula (8) and formula (9) into the following:

$$x = \frac{xC_{TSV}}{xC_{TSV} + (1-x)C_{TSV}} \tag{10}$$

which is the final step in Fig. 11.

We have prepared a simulation to verify the whole test and diagnosis process. We use the open fault data listed in TABLE 3 to model the TSV and the DfT structure shown in Fig. 9 to build the test circuits. The complete circuits are designed using the Nangate 45 nm open cell library and the 45 nm Predictive Technology Model (PTM). The other simulation parameters are listed as follows: $V_{DD} = 1.1\ V$, $V_{TH} = 0.5\ V$, $T_c = 20\ ns$, and C_s is choosed of 10 pF. TABLE 7 shows the simulation results, which include xC_{TSV} and $(1-x)C_{TSV}$ as calculated based on formula (8) and (9).

The first column shows the location of an open fault, the second column shows the value of C_1 in the TSV lumped

TABLE 7. Simulation results of TSV void defect detection and diagnosis.

Void Location	$C_1(fF)$	Measured t (ns)	Calculated xC_{TSV} (fF)	$C_2(fF)$	Measured t' (ns)	Calculated $(1-x)C_{TSV}$ (fF)
10%	5.4	22160	7.1	56.0	2720	57.9
30%	18.4	7820	20.1	43	3520	44.8
50%	30.1	4940	31.9	31.4	4740	33.2
70%	42.4	3560	44.3	18.9	7620	20.7
90%	55.3	2760	57.1	6.0	20340	7.7

TABLE 8. Simulation results of TSV pinhole defect detection and diagnosis.

Pin-hole Size	R_3 (Ω)	R_4 (Ω)	The Practical Leakage Resistance(Ω)	Measured t (ns)	Calculated $R_{leakage}$ (Ω)
$1\ \mu m \times 1\ \mu m$	30396	30396	15198	118	14967
$2\ \mu m \times 2\ \mu m$	13710	13710	6855	51	6468
$3\ \mu m \times 3\ \mu m$	8237	8237	4118.5	27	3424
$4\ \mu m \times 4\ \mu m$	5878	5878	2939	16	2029
$5\ \mu m \times 5\ \mu m$	4291	4291	2145.5	10	1268

defect model, the third column shows the measured discharging time t from die 1, the fourth column shows the calculated xC_{TSV} using the measured t, the fifth column shows the value of C_2 in the TSV lumped defect model, the sixth column shows the measured discharging time t' from die 2, and the seventh column shows the calculated $(1-x)C_{TSV}$ using the measured t'. The calculated xC_{TSV} is very close to C_1 under testing, and the calculated $(1-x)C_{TSV}$ is very close to C_2 under testing. The average error of approximately 1.8 fF arises from the parasitic capacitance of the TSV original drivers and the TGs, which cannot be eliminated. However, the measured results are still high enough to conduct defect diagnosis. According to the ratio between xC_{TSV} and $(1-x)C_{TSV}$, we can determine the location x of open faults by formula (10).

For example, TABLE 7 shows that an open fault occurs at 30%, which has a calculated xC_{TSV} of 20.1 fF and $(1-x)C_{TSV}$ of 44.8 fF. According to formula (10), we calculate the fault location to be $20.1/(20.1+44.8) \approx 31\%$, which is quite near to the practical state.

As a conclusion, by implementing test steps as Fig. 11 shows, the proposed method can detect open fault caused by void defect and diagnose the location of the void.

B. TSV PINHOLE DEFECT DETECTION AND DIAGNOSIS

In section II, TABLE 4 shows that different sizes of pinhole defects will alter the resistance of lumped defect model (R_3, R_4) to different degrees. Since R_3 and R_4 act in parallel, the practical leakage resistance can be calculated as follows:

$$R_{leakage} = \frac{R_3 \cdot R_4}{R_3 + R_4} \tag{11}$$

Therefore, we can measure $R_{leakage}$ to detect and diagnose the pinhole defects. To measure $R_{leakage}$, we reuse the SC circuits and operate the system in a new testing mode. First, we also initialize the system by charging C_s to V_{DD}. Then, by closing TG_1 and opening all other TGs, C_s and $R_{leakage}$ form a first-order discharging circuit. When the TSV is fault-free, no leakage path exists on the dielectric, thus causing the voltage across C_s to remain unchanged. When the TSV contains a pinhole defect, the charge on C_s leaks to the substrate through $R_{leakage}$, thus decreasing the voltage across C_s. The leakage process follows the relationship:

$$V_{CS}(t) = V_{DD} \cdot e^{-\frac{t}{\tau'}} \tag{12}$$

in which $V_{CS}(t)$ is the voltage on C_s and τ' is equal to $C_S \cdot R_{leakage}$.

By recording the discharging time t when $V_{CS}(t)$ drops to a certain threshold V_{TH}, we can calculate $R_{leakage}$ using the following formula:

$$R_{leakage} = \frac{t}{ln(\frac{V_{DD}}{V_{TH}}) \cdot C_s} \tag{13}$$

To verify this testing process, we model the TSV using the data listed in TABLE 6 and use the SC circuit to implement the pinhole defect test. The test results are shown in TABLE 8.

When the pinhole defect is small, the calculated $R_{leakage}$ is close to the practical value. When the pinhole defect is larger, the calculated error becomes larger, primarily because of the nonlinear on-resistance of TG_1, which cannot be eliminated.

This test process can be implemented before die bonding since up to this point we do not need any support of the DfT structure in die 2. Of course, the proposed method can also be used in the postbond stage.

As a conclusion, by implementing test flow as Fig. 12 shows, the proposed method can detect leakage fault caused by pinhole defect and estimate the size of pinhole.

C. TSV MISALIGNED OR MISSING MICRO-BUMP DEFECT DETECTION AND DIAGNOSIS

As we clarify in Section II, TSV misaligned defect does not vary the parameters of the TSV low-bandwidth lumped

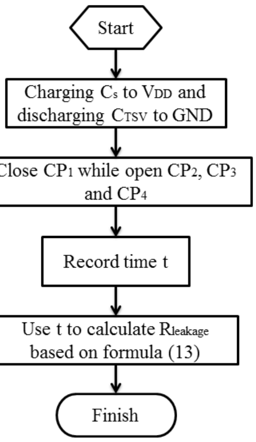

FIGURE 12. **The test flow of TSV pinhole defect detection and diagnosis.**

TABLE 9. **The TSV capacitance measuring resolution under process variation.**

TSV Capacitance (fF)	Calculated C_{TSV} (fF)	Max Absolute Error
54	52 - 56	2 (fF)
56	53 - 58	3 (fF)
58	56 - 61	3 (fF)
60	58 - 62	2 (fF)
62	59 - 64	2 (fF)
64	61 - 67	3 (fF)
66	64 - 68	2 (fF)

TABLE 10. **The TSV leakage resistance measuring resolution under process variation.**

Leakage Resistance ($k\Omega$)	Calculated $R_{Leakage}$ ($k\Omega$)	Max Relative Error
1	1.206 - 1.626	62.6%
10	10.309 - 11.494	14.94%
20	20 - 21.276	6.38%
25	25 - 26.315	5.26%
50	47.619 - 52.631	5.26%
100	100	0%

circuit model, hence, it cannot be detected by the parameter measurement. In this subsection, we focus on detecting the TSV missing micro-bump defect.

We use the TSV low-bandwidth lumped circuit model of the missing micro-bump defect in Section II, Subsection C to build the SC circuits in the HSPICE environment. During the test, the operations of CP signals are the same as in the TSV void defect test. Figure 13 shows the test results.

Clearly, t is 2240 ns when V_{TH} drops below 0.5 V, and the measured t' when V_{TH} drops below 0.5 V is 24040 ns. We insert these data into formula (8) and calculate that $xC_{TSV} = 70.4\,fF$ and $(1-x)C_{TSV} = 6.5\,fF$, which is close to the values of $C_1 = 67\ fF$ and $C_2 = 2.7\ fF$ obtained from the TSV lumped defect model. The error between the calculated value and practical value is induced by the parasitic capacitance of the original drivers and the TGs, which cannot be eliminated.

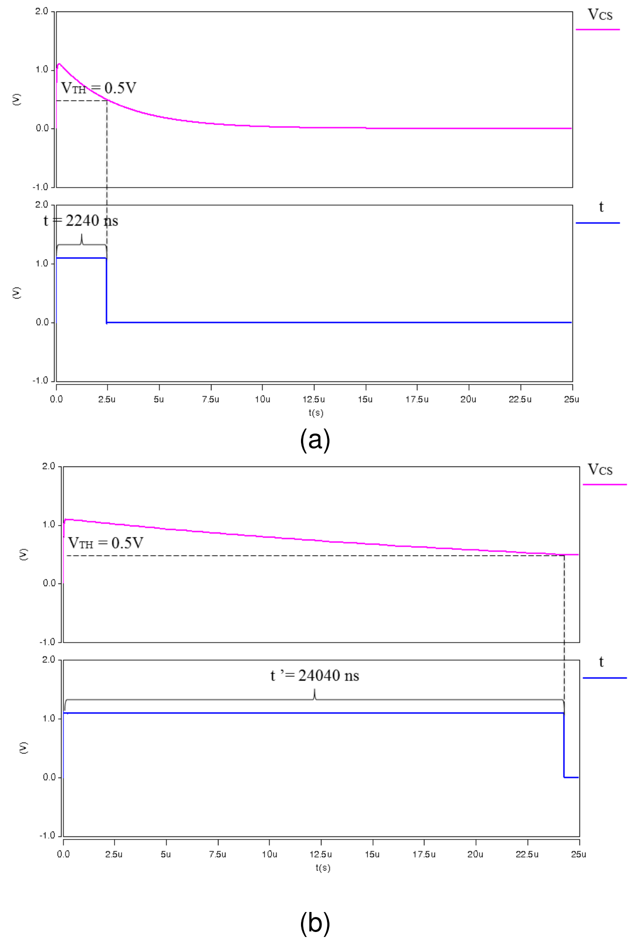

FIGURE 13. **Test results of TSV missing micro-bump defect. (a).t; (b).t'.**

As a conclusion, the proposed method can detect TSV missing micro-bump defects in the post-bond stage. The test flow of TSV missing micro-bump defect is as same as the test flow shown in Fig. 11.

D. FAULT DETECTION AND DIAGNOSIS CAPACITY COMPARED WITH EXISTING TECHNIQUES

In this section, we compare the framework with two typical existing TSV test techniques in the capability of fault detection and diagnosis.

The first typical technique is the Ring Oscillation (RO) based TSV test method [20], [21]. In the post-bond stage, two or more TSVs can form a RO by connecting each other head to tail. Figure 14 shows a simplest RO circuits that consists of two Multiplexers (MUX), four TSV Original Drivers, one Inverter (INV), and two TSVs. The frequency of the RO is influenced by the TSV and the test components along the RO path. When the TSV has an open fault, the RO will not work since the signal path is cut off. When the TSV has a leakage fault, the frequency of RO will decrease. By comparing the captured RO frequency with fault-free case, this technique can detect the existing of open fault and leakage fault.

The second typical technique is the 3D Die Wrappers based post-bond TSV test method [22]. Since 3D die wrappers can

TABLE 11. The comparison in defect detection and diagnosis capacity among two typical TSV test techniques and the proposed method.

Faults (The Causes) \ Different Techniques	The RO based method	The 3D die wrappers based method	The proposed method
Open (Void)	Detectable	Detectable	Detectable and Diagnosable
Leakage (Pinhole)	Detectable and Diagnosable	Detectable	Detectable and Diagnosable
Open (Missing micro-bump)	Detectable	Detectable	Detectable and Diagnosable

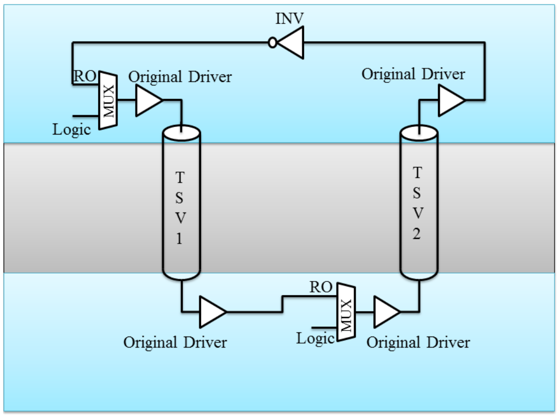

FIGURE 14. A typical RO based TSV post-bond test circuits.

provide full controllability respectively observability at all TSV inputs and outputs, TSV are allowed to be tested as interconnects in post-bond stage. The advantages of 3D die wrappers are standard interface, optional test algorithms, and all-digital design. However, the main drawback is this kind of method can hardly detect weak defect and timing faults.

For the proposed method, we detect TSV defects based on the RC parameters measurement. By using SC technology, the test resolution of the proposed method stays in a relative high level. TABLE 9 and TABLE 10 show the test resolution of the proposed method under process variation of $3\sigma_{V_{th}} = 50\ mV$ and $3\sigma_{Leff} = 10\%$, where $\sigma_{V_{th}}$ is the variation in the threshold voltage and σ_{Leff} is the variation in the CMOS gate length.

TABLE 9 presents that even under process variation, the proposed method can measure TSV capacitance within a absolute error of $3\ fF$. TABLE 10 shows that the proposed method has a high measurement resolution for slight leakage fault, which indicates that the proposed method has a wide coverage in leakage fault detection.

Finally, we compare the above three TSV fault test techniques together in TABLE 11. The first column of TABLE 11 lists the detectable faults and the corresponding causes. The term "Detectable" in TABLE 11 presents the current method can detect the TSV fault whereas the term "Diagnosable" means the current method can give out fault information in detail (the causes, location, degree, etc.).

Compared with the other two methods, the proposed method can not only confirm the TSV faulty or not but also show which defect causes the fault. Moreover, for open fault caused by void defect, the proposed method can provide the location of the void; For leakage fault caused by pinhole defect, the proposed method can estimate the size of the pinhole. All these detail information are valuable since they can be used as feedback for TSV manufacturer to improve the TSV manufacturing technology.

IV. CONCLUSION

In this paper, we focus on TSV void defects, pinhole defects, misaligned defects, and missing micro-bump defects, which are common in 3-D IC manufacturing processes. The low-bandwidth equivalent lumped defect models of the above defects are obtained through a HFSS 3-D full wave simulation. We have analyzed the simulation results and found that only significant void defects, pinhole defects, and missing micro-bump defects can affect the electrical parameters of the lumped circuit model and thus can be detected through parameter measurement. We also present a TSV-defect detection and diagnosis method based on SC circuits. The HSPICE simulation shows that the proposed method can accurately detect void defects, pinhole defects, and missing micro-bump defects in both pre-bond stage and post-bond stage. Moreover, the proposed method can diagnose the location of open faults and the degree of leakage faults in terms of the measured electrical parameters.

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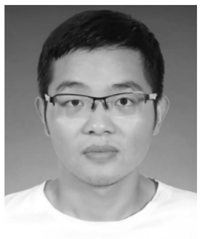

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